

THE +ENERGY FAÇADE

Integrated Building Thermal Energy System

Copa Solar AB × Cosentino Group

Advanced Building-Integrated Renewable Energy Solutions

1. EXECUTIVE SUMMARY

The +Energy Facade system represents a revolutionary integration of Copa Solar's thermal absorber technology (RAM00203) with Cosentino's Dekton® ultra-compact surfaces, creating a comprehensive building-integrated thermal energy solution. This system generates renewable thermal energy directly from building facades while maintaining complete aesthetic integrity and supporting the highest levels of sustainable building certifications.

System Innovation

- | | |
|----------------------------------|---|
| • Silent Operation | Zero moving parts, completely maintenance-free thermal generation |
| • Aesthetic Integration | No visual impact on facade appearance |
| • Dual Functionality | Thermal energy generation with premium surface performance |
| • Carbon Negative Impact | Combined system delivers net negative carbon footprint |
| • Building Certifications | Full support for LEED, BREEAM, and DGNB standards |



TECHNICAL DATASHEET

2. DEKTON® ULTRA-COMPACT SURFACE SPECIFICATIONS

2.1 Product Description

Dekton® is a sophisticated engineered surface manufactured through Particle Sintering Technology (PST), utilizing high-temperature sintering of ultra compacted minerals. The material delivers exceptional performance characteristics specifically suited for facade applications.

Physical Properties

- **Format Options** Up to 3200 × 1440 mm large format slabs
- **Thickness Range** 4 mm, 8 mm, 12 mm, 20 mm, and 30 mm
- **Density** 2,500 - 2,610 kg/m³
- **Service Life** 60 years (Norwegian BREEAM NOR standard)
- **Classification** CPC 373 (Refractory products and structural non-refractory clay products)
Group BI_a (EN 14411:2006)

Performance Characteristics

- **UV Resistance** Exceptional resistance to ultraviolet degradation
- **Scratch Resistance** Superior hardness preventing surface damage
- **Stain Resistance** Non-porous surface with minimal absorption
- **Thermal Shock Resistance** Stable under extreme temperature variations
- **Water Absorption** Very low rates suitable for exterior applications
- **Chemical Composition** 100% inorganic, no resins or organic additives

2.2 Material Composition (per 1000 kg)

Component	Unit	Value	Percentage
Clay	kg	406.8	40.7%
Feldspar	kg	494.2	49.4%
Inorganic Pigments	kg	13.3	1.3%
Other Additives	kg	85.7	8.6%

Sustainability Features

- **Renewable Energy Manufacturing** 100% certified renewable electricity
(64% wind, 25% hydraulic, 11% solar)
- **Recovered Materials** Minimum 5% recovered raw materials incorporation
- **Waste Recovery** 90% production waste minimization through reprocessing
- **Water Management** 95% process water recovery and reuse

TECHNICAL DATASHEET

2.3 LCA Information

- **System Boundaries** Cradle-to-grave and module D (A +B +C + D)
- **Standards** EN 15804:2012+A2:2019, EN ISO 14025:2010
- **Verification** Third-party verified by Marcel Gómez Consultoria Ambiental

2.4 Environmental Information

Core Environmental Impact Indicators (per 1000 kg Dekton®)

Impact Category	Unit	Total Impact	Product Stage (A1-A3)
Climate change - total	kg CO ₂ eq.	867	676
Climate change - fossil	kg CO ₂ eq.	852	671
Climate change -biogenic	kg CO ₂ eq.	6.08	3.35
Ozone Depletion	kg CFC 11 eq.	0.000146	0.0000942
Acidification	mol H ⁺ eq.	4.46	2.06
Eutrophication aquatic freshwater	kg PO ₄ eq.	0.138	0.0842
Eutrophication aquatic marine	kg N eq.	1.08	0.45
Water use	m ³ depriv.	317	277

Use of resources (per 1000 kg Dekton®)

Parameter	Unit	Total Impact	Product Stage (A1-A3)
Total use of renewable primary energy resources	MJ	2,250	2,150
Total use of non-renewable primary energy resources	MJ	14,800	11,200
Use of secondary Material	kg	0	0

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3. COPA SOLAR THERMAL ABSORBER SPECIFICATIONS

3.1 Product Description

RAM002XX is a Building-Integrated Thermal Absorber for Dekton® façade, consisting of aluminium pipes on an aluminium cooling plate. Solar irradiance and ambient air heats the façade slab. The thermal absorber is designed to lead away the excess heat and deliver thermal energy to suitable energy system with fluid circuit design.

Physical Specifications RAM002XX

- **Format Options** Minimum 120 x 120 mm up to 1250 x 2600 mm large format absorbers
- **Assembly with Dekton® facade** Clamp mounting
- **Manufacturing Location** Europe (Germany)
- **Service Life** 25 years with full warranty coverage
- **Classification** CPC 173 (Steam and hot water classification)

Physical Specifications RAM00203

- **Dimensions (L×W×H)** 1100 x 700 x 13 mm (adapted for Dekton® façade 1200x800)
- **Aperture Area** 0.77 m²
- **Weight (empty)** 1.6 kg
- **Weight (filled)** 2.2 kg

Hydraulic Specifications

Parameter	Unit	Value	Notes
Optimal Flow Rate	ℓ/h	150-250	Laminar-turbulent transition at ~193 ℓ/h
Operating Pressure	bar	up to 6	System design pressure
Fluid Temperature Range	°C	-25 to +150	Glycol-water mixture recommended
Hydraulic Connections	-	Push-fit fittings	Compatible with hydronic systems

3.2 Material Characteristics (per m²)

Circularity characteristics	Unit	Value	Notes
Material Composition (plate)	%	100	Aluminium alloy EN AW-1050A
Material Composition (pipe))	%	100	Aluminium alloy EN AW-3103
Recyclability	%	96	Aluminium recycling rate at end-of-life
Secondary Material Content	%	2.48	Recycled content
Rare Earth Metals	%	0	None used in production

Sustainability Features

- **Renewable Energy Manufacturing** 100% certified renewable electricity (solar)
- **Recovered Materials** Minimum 10% recovered raw materials incorporation
- **Waste Recovery** 95% production waste minimization through reprocessing
- **Water Management** 95% process water recovery and reuse

TECHNICAL DATASHEET

3.3 LCA Information

- **System Boundaries** Cradle-to-gate with modules C1-C4 and module D (A1-A3 + C1-C4 + D modules)
- **Standards** EN 15804+A2:2019, EN ISO 14025:2010
- **Verification** Third-party verified by EPD Hub Limited (EPD HUB-1392)

3.4 Environmental Information

Core Environmental Impact Indicators (per m²)

Impact Category	Unit	A1-A3 Product stage	D Module Benefits
Climate change - total	kg CO ₂ eq.	20.0	-10.4
Climate change - fossil	kg CO ₂ eq.	21.1	-10.1
Climate change - biogenic	kg CO ₂ eq.	-1,53	0
Ozone Depletion	kg CFC 11 eq.	0,00000173	-0,00000112
Acidification	mol H ⁺ eq.	0,164	-0,0686
Eutrophication aquatic freshwater	kg PO ₄ eq.	0,000029	-0,000554
Eutrophication aquatic marine	kg N eq.	0,0197	-0,00763
Water use	m ³ depriv.	8.54	-21.2

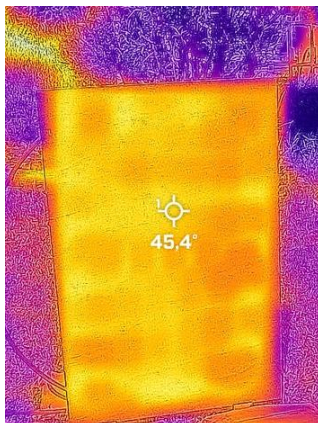
Use of resources (per m²)

Parameter	Unit	Product Stage (A1-A3)	D Module Benefits
Total use of renewable primary energy resources	MJ	148	-81,8
Total use of non-renewable primary energy resources	MJ	249	-159

3.5 Functional Information

- A Dekton® façade slab becomes very warm when exposed to solar irradiation, pic 1.
- Cold thermal fluid is pumped to the Copa Solar Thermal Absorber in a thermal infrastructure system, pic 2.
- The thermal fluid cools the Dekton® façade slab as it transports the heat into the thermal system, pic 3.

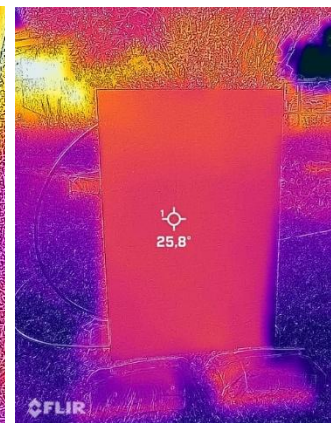
A façade in thermal balance supplies the thermal energy system with heat for hot water and lowers the indoor climate temperature.



1. Front: start temperature



2. Back: start temperature



3. Front: balanced temperature

4. INTEGRATED SYSTEM PERFORMANCE

4.1 Product Description “The +Energy Façade”

The Cosentino Dekton® Façade Slab combined with the Copa Solar Thermal Absorber, is an energy powerhouse, for e.g. warm water production and thermal energy, stored for use when heat is needed.

The need for active cooling inside the building is reduced with The +Energy Façade installed, as the heat in the slab is reduced and transported away from the building's envelope.

The size per +Energy Façade slab has a range between 0.25 and 4.6 m².

Physical Specifications EFA002XX

- **Format Options** Up to 3200 × 1440 mm large format slabs
- **Thickness Range Dekton®** 8 mm, 12 mm, 20 mm, and 30 mm
- **Thickness Copa Solar absorber** 13 mm
- **Combined thickness +Energy Façade** 21 mm, 25 mm, 33 mm, and 43 mm (incl absorber)
- **Assembly** Clamp mounting
- **Manufacturing Location** Europe (Germany and Spain)
- **Service Life** 25 years with full warranty coverage
- **Combined weight 8mm Dekton®** 26,1 – 27,2 kg/m²

Physical Specifications EFA00101

- **Dimensions (L×W×H)** 1200 × 800 × 21 mm
- **Aperture Area** 0.77 m²
- **Weight (empty)** 20,8 kg
- **Weight (filled)** 21,4 kg

Performance Characteristics

- **UV Resistance** Exceptional resistance to ultraviolet degradation
- **Scratch Resistance** Superior hardness preventing surface damage
- **Stain Resistance** Non-porous surface with minimal absorption
- **Thermal Shock Resistance** Stable under extreme temperature variations
- **Water Absorption** Very low rates suitable for exterior applications
- **Chemical Composition** 100% inorganic, no resins or organic additives

Thermal Performance Characteristics

Parameter	Unit	Value	Test Conditions
Installed Thermal Peak Power	W/m ²	1,400	T _{amb} = 45°C, T _{fluid} = 5°C
Radiative Cooling Capacity	W/m ²	200	Night-time operation

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4.2 Combined Environmental Benefits

Carbon Impact Assessment

- Dekton® Facade Slab (1000 kg) 867 kg CO₂ eq total impact
- Thermal Absorber (per m²) 20.0 kg CO₂e production -10.4 kg CO₂e benefits
- Net System Impact The combination delivers significant carbon savings through thermal energy generation that offsets production impacts over the service life.

4.3 Life Cycle Synergies

Service Life Alignment

- Dekton® Design Life 60 years (BREEAM NOR standard)
- Copa Solar Warranty 25 years thermal absorber
- System Optimization Thermal liquid replacement at mid-life maintains peak performance

Maintenance Benefits

- Dekton® Maintenance Minimal cleaning with neutral soap and water
- Copa Solar Maintenance Zero operational maintenance (no moving parts)
- Combined System Maintenance-free thermal generation with easy-clean surfaces

4.4 Energy Performance

Annual Thermal Yield

Location Example mean thermal fluid temperature 5°C

- Stockholm, Sweden 1,480 kWh/m²/year
- Würzburg, Germany 1,660 kWh/m²/year

System Efficiency Benefits

Application	Energy Cost Reduction
Hot Water Production	>30%
Space Heating	>40%
Space Cooling	>35%
Cooling Demand Reduction	>40%

5. BUILDING APPLICATIONS

5.1 Building Integration Types

New Construction Applications

- | | |
|-----------------------------|---|
| • Curtain Wall Integration | Direct mounting to structural glazing systems |
| • Precast Panel Integration | Factory integration with concrete facade panels |
| • Composite Panel Systems | Integration with ACM and metal composite panels |
| • Natural Stone Facades | Combination with stone veneer systems |

Retrofit Applications

- | | |
|-----------------------------|--|
| • Existing Facade Overlay | Adhesive mounting to existing surfaces |
| • EIFS Integration | Combination with external insulation systems |
| • Cladding Replacement | Upgrade of existing facade materials |
| • Partial Facade Renovation | Strategic placement for optimal solar exposure |

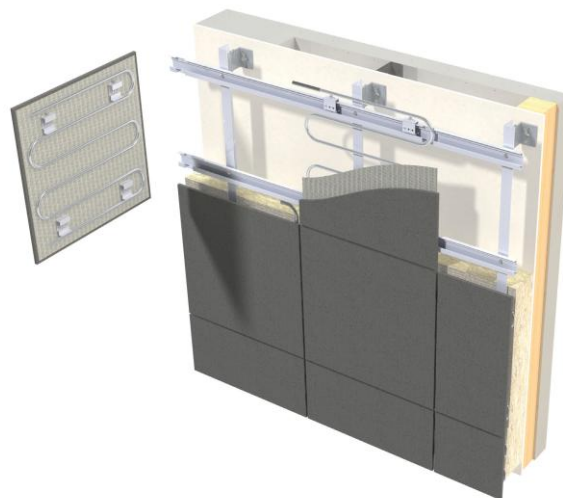
5.2 Building Performance Integration

HVAC System Coupling

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|-------------------------|--|
| • Heat Pump Integration | Improved COP through consistent heat source |
| • Hydronic Distribution | Direct connection to radiant heating/cooling systems |
| • Thermal Storage | Integration with seasonal energy storage systems |
| • Smart Controls | Building automation system integration |

Energy System Benefits

- | | |
|-----------------------|--|
| • Peak Load Reduction | Thermal generation reduces conventional energy demand |
| • Grid Independence | Reduced reliance on utility thermal energy |
| • Demand Response | Thermal storage enables grid balancing participation |
| • Energy Resilience | On-site thermal generation improves building energy security |
| • Payback Period | 3-5 years depending on price of replaced energy |



6. CERTIFICATIONS AND STANDARDS

6.1 Green Building Standards Support

LEED v4.1 Contributions

Energy & Atmosphere Credits

- EA Credit 2 Optimize Energy Performance (thermal generation contribution)
- EA Credit 3 Advanced Energy Metering (system monitoring capability)

Materials & Resources Credits

- MR Credit 1 Building Product Disclosure (EPD documentation)
- MR Credit 2 Environmental Product Declarations (verified EPDs)

Innovation Credits

- IN Credit 1 Innovation (integrated facade thermal system)

BREEAM International Support

Energy Category

- ENE 01 Energy Efficiency (thermal system contribution)
- ENE 02 Energy Monitoring (system performance tracking)

Materials Category

- MAT 01 Life Cycle Impacts (verified LCA data)
- MAT 03 Responsible Sourcing (certified sustainable materials)

Innovation Category

- INN 01 Innovation (novel integrated approach)

DGNB System Qualification

Technical Quality

- TEC 1.1 Building Life Cycle Costs (60-year service life)
- TEC 3.1 Use and Maintenance Friendliness (minimal maintenance)

Environmental Quality

- ENV 1.1 Life Cycle Assessment (comprehensive LCA documentation)
- ENV 2.1 Local Environmental Impact (minimal local emissions)

6.2 Compliance Documentation

Standards Compliance

Dekton® Certifications

- ISO 9001:2015 Quality Management Systems
- ISO 14001:2015 Environmental Management Systems
- ISO 20400:2017 Sustainable Procurement
- UNE 15896:2015 Value Added Purchasing Management

Copa Solar Certifications

- ISO 9001:2015 Quality Management Systems
- EN 12975:2022 Solar collectors

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7. INSTALLATION & TECHNICAL SPECIFICATIONS

7.1 Installation Requirements

Facade Mounting System

Dekton® Installation

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|-----------------------|---|
| • Structural Support | Engineered mounting systems for large format panels |
| • Thermal Movement | Expansion joint accommodation |
| • Weather Sealing | Comprehensive weatherproofing systems |
| • Access Requirements | Service access for thermal system maintenance |

Copa Solar Integration

- | | |
|---------------------------|---|
| • Attachment Method | Clamping |
| • Substrate Compatibility | Direct mounting to Dekton® surface |
| • Installation Time | <30 minutes per thermal absorber unit |
| • System Integration | Standard compression fittings for fluid connections |

Fluid Circuit Design

System Requirements

- | | |
|----------------------|--|
| • Recommended Fluid | Propylene glycol-water mixture (frost protection) |
| • Flow Rate | 150-250 ℓ/h per absorber unit |
| • Operating Pressure | 2-4 bar system pressure |
| • Distribution | Piping with freeze protection |
| • Controls | Automatic circulation control with temperature sensors |

7.2 Performance Monitoring

System Monitoring Capabilities

- | | |
|-------------------------|---|
| • Thermal Output | Real-time thermal energy generation measurement |
| • Fluid Temperatures | Supply and return temperature monitoring |
| • Flow Rates | Hydraulic performance verification |
| • Energy Balance | Building energy management system integration |
| • Performance Analytics | Long-term performance trend analysis |

Maintenance Protocols

Dekton® Surface Maintenance

- | | |
|----------------------|---|
| • Cleaning Schedule | Weekly cleaning recommended (50 cleanings/year) |
| • Cleaning Method | Neutral soap and water (0.2 kg soap/year) |
| • Visual Inspection | Annual condition assessment |
| • Surface Protection | UV and weather resistance monitoring |

Copa Solar System Maintenance

- | | |
|----------------------------|--|
| • Operational Maintenance | None required (no moving parts) |
| • Fluid System | Standard hydronic system maintenance |
| • Performance Verification | Annual thermal output testing |
| • Component Inspection | Visual check of mounting and connections |

8. QUALITY ASSURANCE & WARRANTIES

8.1 Product Warranties

Dekton® Warranty Coverage

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|-------------------------|--|
| • Material Warranty | Comprehensive coverage against manufacturing defects |
| • Performance Warranty | UV resistance, stain resistance, thermal performance |
| • Installation Warranty | Professional installation guarantee |
| • Service Life | 60-year design life with proper maintenance |

Copa Solar Warranty Coverage

- | | |
|-------------------------|-------------------------------------|
| • Performance Warranty | 25 years thermal output guarantee |
| • Product Warranty | Manufacturing defect coverage |
| • Installation Warranty | Professional installation guarantee |
| • Component Warranty | Individual component coverage terms |

8.2 Quality Control Systems

Manufacturing Quality

Dekton® Quality Systems

- | | |
|-----------------------|--|
| • ISO 9001:2015 | Quality management systems |
| • ISO 14001:2015 | Environmental management systems |
| • Process Control | Automated production monitoring |
| • Material Testing | Comprehensive material property verification |
| • Third-Party Testing | Independent performance validation |

Copa Solar Quality Systems:

- | | |
|--------------------------|---|
| • ISO 9001:2015 | Quality management systems |
| • ISO 14001:2015 | Environmental management systems (according to) |
| • EN 12975:2022 | Solar collectors |
| • Process Control | Automated production monitoring |
| • Material Testing | Comprehensive material property verification |
| • Third-Party Testing | Independent performance validation |
| • Performance Validation | Independent testing verification |

TECHNICAL DATASHEET

9. CONTACT INFORMATION & TECHNICAL SUPPORT

9.1 Primary Contacts

Copa Solar AB

- Specializations:**
- Solar thermal systems design and engineering
 - Thermal absorber technology development
 - Energy consulting and system optimization
 - OEM solutions and custom applications



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Cosentino Group

- Specializations:**
- Dekton® ultra-compact surfaces
 - Sustainable building materials development
 - Architectural design solutions
 - Global distribution and technical support



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9.2 Technical Support Services

Engineering Support

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|--------------------------------|---|
| • System Design | Integrated facade thermal system design |
| • Performance Modelling | Energy generation and building performance analysis |
| • Installation Guidance | Technical installation support and training |
| • Commissioning | System startup and performance verification |

Documentation Support

- | | |
|-----------------------------------|--|
| • Technical Specifications | Detailed product and system documentation |
| • Certification Support | Green building certification assistance |
| • Compliance Documentation | Environmental and performance verification |
| • Warranty Administration | Product warranty and service support |



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10. APPENDICES

Appendix A: Technical Standards References

- | | |
|-----------------------------|---------------------------------------|
| • UNE-EN 15804:2012+A2:2019 | Sustainability of construction works |
| • ISO 14025:2006 | Environmental labels and declarations |
| • EN 12975:2022 | Solar collectors |
| • EN 14411:2006 | Ceramic tiles classification |
| • ISO 9001:2015 | Quality management systems |
| • ISO 14001:2015 | Environmental management systems |

Appendix B: Environmental Assessment Methodologies

- | | |
|--------------------------------------|---------------------------|
| • Life Cycle Assessment | ISO 14040/14044 standards |
| • Environmental Product Declarations | EN 15804+A2:2019 |
| • Water Footprint | AWARE methodology (2016) |
| • Cumulative Energy Demand | CED methodology v1.11 |
| • Impact Assessment | EF 3.0 Method |

Appendix C: Building Code Compliance

- | | |
|---|--|
| • International Building Code (IBC) | Facade and structural requirements |
| • International Energy Conservation Code (IECC) | Energy efficiency compliance |
| • European Construction Products Regulation (CPR) | CE marking requirements |
| • National Building Codes | Country-specific compliance verification |

Document Information:

- | | |
|--------------------------|---|
| • Version: | 1.0 |
| • Date: | 2025-03-17 |
| • Classification: | Technical Documentation - Integrated System |
| • Based on: | Verified third-party testing, environmental assessments, and certified product data |
| • Status: | Specifications subject to continuous improvement and technical modifications |

Quality Assurance: This documentation integrates verified data from third-party Environmental Product Declarations, certified test results, and comprehensive life cycle assessments to provide accurate technical and environmental performance information for the integrated +Energy Façade combining Dekton® with Copa Solar Thermal Absorber